

Sebastian Ashkar

sebastian11ryan@gmail.com | sebastianashkar.com | linkedin.com/in/sebastianashkar | github.com/AwesomeDemoGuy

EDUCATION

Carnegie Mellon University

Master of Science in Information Security

Expected Dec 2028

Arizona State University

Bachelor of Science in Computer Science (Cybersecurity), Minor in Political Science

Expected Dec. 2026

- Coursework: Advanced Web Security, Computer Architecture, Computer Forensics, Computer Networks, Design and Analysis of Algorithms, Systems Security
- pwn.college Yellow Belt, New American University - Provost Award Scholar, 6x Dean's List, 3.99 GPA.
- CSAW CTF '26 quals 5th place in USA

EXPERIENCE

Software Engineering Intern

May 2025 – Aug. 2025

Cisco

San Jose, CA

- Drove the implementation of a data visualization web dashboard utilizing Plotly Dash.
- Leveraged GitHub Copilot to accelerate development, contributing 20+ unique features and 30k+ lines of production code.
- Implemented and optimized data gathering and ETL workflows to and from various MongoDB and PostgreSQL databases.
- Achieved millisecond data fetch time. Reduced page load time by 10x.
- Mitigated SQL injection and XSS vulnerabilities through input filtering.
- Collaborated with a talented team of back-end and front-end developers to ensure optimal performance and delivery time.
- Successfully presented product and contributions to panel of senior directors.

Undergraduate Teaching Assistant

Aug. 2026 – Dec. 2026

Arizona State University

Tempe, AZ

- Held weekly office hours and debugging sessions to stimulate an in-depth understanding of Cybersecurity and Networking.
- Computer Networks: Mentored students through hands-on networking labs, providing technical support for socket programming, packet analysis (Wireshark), TCP/IP protocols, and routing algorithms.
- Information Assurance: Scaled support across a 600+ student cohort by troubleshooting CTF challenges, clarifying exploits (e.g., buffer overflows, machine-in-the-middle, web security, cryptography, reverse engineering), and clarifying core concepts.

PROJECTS

Loncapa Solver | *Python, Parser, Discord API, Git* | [GitHub](#) | [Top.gg](#)

- Collaborated on an extensive multi-year mathematical problem-solving AI Discord bot to aid students in mastering physics.
- Leveraged Discord.py library to streamline user interactions and Stripe API to implement secure payment processing.
- Efficiently parsed user input and isolated key values by calculating the Levenshtein distance between strings.
- 4k+ lines of code contributed, over 500 physicists aided, 23k+ requests served.

Sarcasm Detector | *TensorFlow, Neural Networks* | [GitHub](#)

- Constructed and trained a custom convolutional neural network that identified sarcasm in newspaper headlines.
- Optimized the node structure to achieve a 75% accuracy in headline identification.

File Compressor-Decompressor | *Java, Git, BST, OOP, Huffman Coding* | [GitHub](#)

- Led a team of four junior developers in developing a file compression and decompression program.
- Utilized the Huffman coding algorithm to reduce the bit size of each character in a text file by upwards of 30%.
- Supported teammates in attaining a deep comprehension of the collaborative coding environment through git.

Twitter Eliminator | *Asyncio, Git, Hikari, Torch, Transformers* | [GitHub](#) | [Top.gg](#)

- Deleted Twitter links, links to Twitter embedding sites, and screenshots of Twitter sent in Discord channels.
- Scanned all image media using a custom ML model to identify and delete screenshots of Twitter.

TECHNICAL SKILLS

Languages: C/C++, Java, MongoDB Script, Python, SQL, JavaScript, HTML/CSS, R, Bash Script, X86 Assembly

Technologies: Apache Airflow, Binary Exploitation, Copilot, Cryptography, GDB, IDA Decompiler, ROP, SSL/TLS

Libraries: Discord.py, Flask, NumPy, Pandas, Plotly, PyCrypto, Dash, Pwntools, Scapy, Stripe